

**KHOSEDA-KHARDSKY, Russian Federation**

WMO: 232190

Lat: **67.033N**Lon: **59.400E**Elev: **83**StdP: **100.33**Time Zone: **3.00 (E03)**Period: **94-16**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB   |              | Humidification DP/MCDB and HR |            |              |              |            |              | Coldest Month WS/MCDB |              |             |              | MCWS/PCWD to 99.6% DB |          | WSF          |
|---------------|--------------|--------------|-------------------------------|------------|--------------|--------------|------------|--------------|-----------------------|--------------|-------------|--------------|-----------------------|----------|--------------|
|               |              |              | 99.6%                         |            |              | 99%          |            |              | 0.4%                  |              | 1%          |              |                       |          |              |
|               | 99.6%        | 99%          | DP                            | HR         | MCDB         | DP           | HR         | MCDB         | WS                    | MCDB         | WS          | MCDB         | MCWS                  | PCWD     |              |
| (a)<br>1      | (b)<br>-42.9 | (c)<br>-39.9 | (d)<br>-44.8                  | (e)<br>0.0 | (f)<br>-42.2 | (g)<br>-42.2 | (h)<br>0.1 | (i)<br>-39.5 | (j)<br>15.0           | (k)<br>-14.6 | (l)<br>13.5 | (m)<br>-12.5 | (n)<br>0.7            | (o)<br>0 | (p)<br>0.823 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.4                   | 26.1            | 18.7 | 23.9 | 17.4 | 21.2 | 16.0 | 19.7                | 25.2 | 18.2 | 22.7 | 16.6 | 20.4 | 3.2                  | 180  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 17.5                            | 12.7 | 22.7 | 16.1 | 11.6 | 21.1 | 14.7 | 10.5 | 19.2 | 56.6          | 25.4 | 51.9 | 23.1 | 46.9 | 20.3 | 23.1              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 11.0              | 9.6  | 8.3 | N/A                        | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A  | N/A        | N/A  | N/A        | N/A  |
|                   |      |     | -44.0                      | 21.7 | 3.0                | 1.0 | -46.2                                              | 22.5 | -47.9      | 23.1 | -49.5      | 23.7 | -51.7      | 24.4 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

**Monthly Climatic Design Conditions**

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |       |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |       |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | -4.5  | -21.4 | -19.5 | -14.3 | -6.5  | 0.9   | 9.5   | 13.9  | 9.4   | 4.7   | -2.5  | -12.3 | -16.7 | (6)  |
| (7)  |                                                                                  | DBStd       | 14.04 | 10.09 | 10.48 | 8.70  | 7.45  | 6.32  | 5.51  | 4.91  | 3.47  | 4.27  | 5.98  | 9.55  | 10.67 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 5538  | 973   | 827   | 754   | 497   | 290   | 76    | 18    | 54    | 166   | 388   | 668   | 827   | (8)  |
| (9)  |                                                                                  | HDD18.3     | 8351  | 1232  | 1060  | 1013  | 747   | 542   | 268   | 154   | 279   | 408   | 647   | 918   | 1085  | (9)  |
| (10) |                                                                                  | CDD10.0     | 248   | 0     | 0     | 0     | 0     | 7     | 60    | 138   | 34    | 8     | 0     | 0     | 0     | (10) |
| (11) |                                                                                  | CDD18.3     | 18    | 0     | 0     | 0     | 0     | 0     | 2     | 16    | 1     | 0     | 0     | 0     | 0     | (11) |
| (12) |                                                                                  | CDH23.3     | 252   | 0     | 0     | 0     | 0     | 4     | 30    | 208   | 11    | 0     | 0     | 0     | 0     | (12) |
| (13) |                                                                                  | CDH26.7     | 50    | 0     | 0     | 0     | 0     | 0     | 3     | 46    | 2     | 0     | 0     | 0     | 0     | (13) |
| (14) | Wind                                                                             | WSAvg       | 3.7   | 3.6   | 3.6   | 3.8   | 3.9   | 4.0   | 3.9   | 3.4   | 3.5   | 3.3   | 3.7   | 3.6   | 4.1   | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 455   | 30    | 23    | 21    | 21    | 32    | 41    | 54    | 66    | 51    | 49    | 34    | 30    | (15) |
| (16) |                                                                                  | PrecMax     | 655   | 59    | 45    | 39    | 51    | 54    | 98    | 160   | 153   | 116   | 90    | 56    | 63    | (16) |
| (17) |                                                                                  | PrecMin     | 290   | 13    | 0     | 5     | 3     | 1     | 10    | 2     | 14    | 1     | 22    | 15    | 10    | (17) |
| (18) |                                                                                  | PrecStd     | 90    | 12    | 10    | 8     | 11    | 13    | 20    | 31    | 28    | 24    | 18    | 12    | 12    | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | -1.3  | 0.4   | 4.0   | 10.7  | 23.6  | 26.0  | 30.1  | 24.6  | 19.4  | 9.7   | 2.4   | 0.7   | (19) |
| (20) |                                                                                  |             | MCWB  | -2.0  | -0.2  | 2.0   | 6.9   | 16.0  | 18.2  | 20.2  | 18.3  | 14.6  | 8.7   | 2.1   | 0.4   | (20) |
| (21) |                                                                                  | 2%          | DB    | -3.0  | -1.3  | 1.0   | 6.2   | 17.9  | 24.0  | 27.7  | 21.1  | 15.8  | 7.4   | 0.7   | -0.2  | (21) |
| (22) |                                                                                  |             | MCWB  | -3.6  | -1.9  | 0.0   | 3.6   | 11.8  | 17.4  | 19.7  | 15.0  | 12.3  | 6.7   | 0.3   | -0.6  | (22) |
| (23) |                                                                                  | 5%          | DB    | -4.6  | -2.9  | -0.1  | 4.3   | 13.5  | 21.5  | 25.6  | 18.4  | 13.4  | 5.6   | 0.1   | -1.8  | (23) |
| (24) |                                                                                  |             | MCWB  | -5.1  | -3.5  | -0.7  | 2.4   | 9.4   | 16.3  | 18.7  | 14.3  | 11.0  | 5.1   | -0.2  | -2.3  | (24) |
| (25) |                                                                                  | 10%         | DB    | -8.0  | -5.1  | -2.0  | 2.7   | 9.9   | 19.2  | 23.2  | 16.2  | 11.5  | 4.4   | -1.0  | -3.5  | (25) |
| (26) |                                                                                  |             | MCWB  | -8.5  | -5.6  | -2.8  | 1.4   | 7.1   | 14.6  | 17.7  | 13.3  | 9.7   | 4.0   | -1.4  | -4.0  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | -1.6  | -0.1  | 2.3   | 7.0   | 16.2  | 20.3  | 21.6  | 19.1  | 15.9  | 8.9   | 2.1   | 0.6   | (27) |
| (28) |                                                                                  |             | MCDWB | -1.0  | 0.3   | 3.8   | 10.6  | 22.7  | 24.6  | 26.9  | 22.6  | 18.6  | 9.6   | 2.5   | 0.6   | (28) |
| (29) |                                                                                  | 2%          | WB    | -3.6  | -1.9  | 0.2   | 4.0   | 12.5  | 18.3  | 20.4  | 16.1  | 13.2  | 6.8   | 0.5   | -0.6  | (29) |
| (30) |                                                                                  |             | MCDWB | -3.1  | -1.5  | 0.8   | 5.6   | 17.6  | 22.3  | 26.5  | 19.3  | 15.0  | 7.2   | 0.6   | -0.3  | (30) |
| (31) |                                                                                  | 5%          | WB    | -5.2  | -3.5  | -0.7  | 2.6   | 9.7   | 16.7  | 19.3  | 14.7  | 11.2  | 5.2   | -0.2  | -2.3  | (31) |
| (32) |                                                                                  |             | MCDWB | -4.5  | -3.0  | -0.1  | 4.0   | 13.4  | 20.9  | 24.9  | 17.8  | 12.9  | 5.5   | 0.0   | -1.9  | (32) |
| (33) |                                                                                  | 10%         | WB    | -8.5  | -5.6  | -2.7  | 1.5   | 7.2   | 15.1  | 17.9  | 13.3  | 9.8   | 4.0   | -1.5  | -4.0  | (33) |
| (34) |                                                                                  |             | MCDWB | -8.0  | -5.1  | -2.1  | 2.5   | 9.9   | 19.2  | 22.5  | 16.0  | 11.3  | 4.4   | -1.1  | -3.5  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | MDBR        |       | 8.7   | 8.4   | 11.1  | 9.4   | 8.1   | 9.3   | 10.4  | 8.1   | 7.1   | 5.2   | 6.9   | 7.6   | (35) |
| (36) |                                                                                  | 5% DB       | MCDBR | 7.4   | 5.9   | 7.6   | 8.9   | 14.5  | 13.2  | 14.6  | 13.4  | 10.2  | 5.0   | 4.5   | 6.0   | (36) |
| (37) |                                                                                  |             | MCWBR | 7.3   | 5.9   | 7.0   | 7.3   | 10.0  | 8.3   | 7.7   | 8.4   | 7.1   | 4.7   | 4.6   | 6.1   | (37) |
| (38) |                                                                                  | 5% WB       | MCDBR | 7.6   | 6.1   | 7.3   | 7.9   | 13.8  | 12.6  | 13.2  | 12.0  | 8.7   | 4.8   | 4.9   | 6.4   | (38) |
| (39) |                                                                                  |             | MCWBR | 7.5   | 6.1   | 6.9   | 6.7   | 9.8   | 8.8   | 7.8   | 8.3   | 6.5   | 4.7   | 4.9   | 6.5   | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        |       | 0.215 | 0.265 | 0.304 | 0.332 | 0.345 | 0.355 | 0.386 | 0.365 | 0.323 | 0.277 | 0.207 | N/A   | (40) |
| (41) |                                                                                  | taud        |       | 1.885 | 2.014 | 2.060 | 2.088 | 2.228 | 2.432 | 2.355 | 2.395 | 2.467 | 2.228 | 1.925 | N/A   | (41) |
| (42) |                                                                                  | Ebn at Noon |       | 133   | 536   | 722   | 810   | 846   | 851   | 803   | 783   | 735   | 538   | 130   | 0     | (42) |
| (43) |                                                                                  | Edh at Noon |       | 14    | 56    | 92    | 118   | 115   | 97    | 102   | 88    | 64    | 45    | 12    | 0     | (43) |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      |       | 0.05  | 0.51  | 2.01  | 3.86  | 5.08  | 5.00  | 4.91  | 3.07  | 1.72  | 0.55  | 0.11  | 0.00  | (44) |
| (45) |                                                                                  | RadStd      |       | 0.01  | 0.09  | 0.28  | 0.35  | 0.49  | 0.52  | 0.77  | 0.35  | 0.20  | 0.09  | 0.01  | 0.00  | (45) |

**Historical Trends**

|                             |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only           |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (2 neighbors) |  | +1.28 | N/A     | N/A    | N/A     | N/A   | N/A   | -406        | -448    | N/A     | N/A     |

Nomenclature: See separate page